

The Five Rules of Conditional Independence

and Their Converses

Let X, Y, Z, W be disjoint sets of random variables. Write $X \perp Y \mid Z$ for

$$p(X, Y \mid Z) = p(X \mid Z) p(Y \mid Z), \quad \forall Z : p(Z) > 0,$$

equivalently $p(X \mid Y, Z) = p(X \mid Z)$. These are the **semi-graphoid / graphoid axioms** governing how conditional independence statements combine.

1 Statement of the Five Rules

Each rule is stated as a one-directional implication; Section 3 addresses whether each converse also holds.

The Five Rules

1. **Symmetry:** $X \perp Y \mid Z \implies Y \perp X \mid Z$.
2. **Decomposition:** $X \perp (Y, W) \mid Z \implies X \perp Y \mid Z$ and $X \perp W \mid Z$.
3. **Weak Union:** $X \perp (Y, W) \mid Z \implies X \perp Y \mid (W, Z)$ and $X \perp W \mid (Y, Z)$.
4. **Contraction:** $X \perp Y \mid (W, Z)$ and $X \perp W \mid Z \implies X \perp (Y, W) \mid Z$.
5. **Intersection** (requires $p(x) > 0$): $X \perp Y \mid (W, Z)$ and $X \perp W \mid (Y, Z) \implies X \perp (Y, W) \mid Z$.

Positivity: only Intersection needs the strict-positivity assumption $p(x) > 0$ for all x . Rules 1–4 hold for *any* probability distribution.

2 Proofs of the Five Rules

Proof 1: Symmetry

By the product rule, $p(Y \mid X, Z) = \frac{p(X, Y \mid Z)}{p(X \mid Z)}$ for all $p(X \mid Z) > 0$. Applying the hypothesis $X \perp Y \mid Z$, i.e. $p(X, Y \mid Z) = p(X \mid Z)p(Y \mid Z)$:

$$p(Y \mid X, Z) = \frac{p(X \mid Z) p(Y \mid Z)}{p(X \mid Z)} = p(Y \mid Z).$$

Hence $Y \perp X \mid Z$. ■

Proof 2: Decomposition

By hypothesis, $p(X, Y, W \mid Z) = p(X \mid Z) p(Y, W \mid Z)$. Marginalizing both sides over W :

$$p(X, Y \mid Z) = \int p(X, Y, W \mid Z) dW = \int p(X \mid Z) p(Y, W \mid Z) dW = p(X \mid Z) p(Y \mid Z),$$

so $X \perp Y \mid Z$. Marginalizing over Y instead gives $X \perp W \mid Z$ by the same argument. ■

Proof 3: Weak Union

$$\begin{aligned}
 p(X, Y \mid W, Z) &= \frac{p(X, Y, W \mid Z)}{p(W \mid Z)} && \text{(product rule)} \\
 &= \frac{p(X \mid Z) p(Y, W \mid Z)}{p(W \mid Z)} && \text{(hypothesis: } X \perp (Y, W) \mid Z) \\
 &= p(X \mid Z) p(Y \mid W, Z) && \text{(product rule)} \\
 &= p(X \mid W, Z) p(Y \mid W, Z) && \text{(Decomposition: } X \perp (Y, W) \mid Z \Rightarrow X \perp W \mid Z)
 \end{aligned}$$

Hence $X \perp Y \mid (W, Z)$. Swapping the roles of Y and W throughout gives $X \perp W \mid (Y, Z)$ by the identical argument. ■

On the condition $p(W \mid Z) > 0$: the first step divides by $p(W \mid Z)$, which requires $p(W \mid Z) > 0$ — but this is *not* an extra assumption beyond ordinary positivity.

- By the definition at the start of this document, every conditional independence statement $X \perp Y \mid Z$ is implicitly asserted only $\forall Z : p(Z) > 0$ — conditioning on a zero-probability event is undefined, so it is excluded from the claim rather than assumed away.
- The conclusion $X \perp Y \mid (W, Z)$ is likewise only a claim about (w, z) pairs with $p(w, z) > 0$, and that is exactly the same set of pairs for which the proof's division by $p(W \mid Z)$ is valid.
- The proof's domain of validity therefore matches the conclusion's domain of validity automatically, with no additional hypothesis required — unlike Intersection below, which needs positivity of the *full* joint distribution, not just of the variables being conditioned on (see the remark after Proof 5).

Proof 4: Contraction

By hypothesis $X \perp Y \mid (W, Z)$, so $p(X \mid Y, W, Z) = p(X \mid W, Z)$. By hypothesis $X \perp W \mid Z$, so $p(X \mid W, Z) = p(X \mid Z)$. Chaining these:

$$p(X \mid Y, W, Z) = p(X \mid W, Z) = p(X \mid Z).$$

Multiplying through by $p(Y, W \mid Z)$ and applying the product rule $p(X, Y, W \mid Z) = p(X \mid Y, W, Z) p(Y, W \mid Z)$:

$$p(X, Y, W \mid Z) = p(X \mid Y, W, Z) p(Y, W \mid Z) = p(X \mid Z) p(Y, W \mid Z),$$

which is exactly $X \perp (Y, W) \mid Z$. ■

Proof 5: Intersection (requires $p(x) > 0$)

From the two hypotheses:

$$X \perp Y \mid (W, Z) \implies p(X \mid W, Y, Z) = p(X \mid W, Z),$$

$$X \perp W \mid (Y, Z) \implies p(X \mid W, Y, Z) = p(X \mid Y, Z),$$

so $p(X \mid W, Z) = p(X \mid Y, Z)$. Now marginalize over W :

$$\begin{aligned} p(X \mid Z) &= \int p(X, W \mid Z) dW && \text{(marginalization)} \\ &= \int p(X \mid W, Z) p(W \mid Z) dW && \text{(product rule)} \\ &= p(X \mid Y, Z) \int p(W \mid Z) dW && \text{(the equality above, valid for every } W) \\ &= p(X \mid Y, Z). \end{aligned}$$

Therefore

$$p(X \mid W, Y, Z) = p(X \mid W, Z) = p(X \mid Y, Z) = p(X \mid Z) \implies X \perp (Y, W) \mid Z. \blacksquare$$

Where positivity is used:

- $p(X \mid W, Z) = p(X \mid Y, Z)$ was obtained by chaining two equalities that individually hold only at (w, y, z) triples with $p(w, y, z) > 0$.
- The next step treats this equality as holding for *every* W with $p(W \mid Z) > 0$ (for the fixed value of Y), so that $p(X \mid Y, Z)$ can be pulled outside the integral over W .
- This is licensed only if every (w, y, z) triple has positive joint probability — i.e. if $p(x) > 0$ everywhere.
- If some triple has zero probability (as in the XOR example of Section 3), this chaining step is not licensed and the conclusion can fail.

Why Weak Union Needs No Positivity Assumption but Intersection Does

Both proofs divide by a conditional probability at some point, so it is worth being precise about why only one of them needs an extra hypothesis.

Weak Union's $p(W \mid Z) > 0$ is *local* and automatic.

- It is exactly the same condition under which the conclusion $X \perp Y \mid (W, Z)$ is even meaningful to state: conditioning on (w, z) is undefined unless $p(w, z) > 0$.
- So the claim is, by the definition on page 1, only ever made at such (w, z) in the first place.
- The proof's domain of validity and the conclusion's domain of validity therefore coincide *by construction*.
- No assumption beyond “the conditioning event has positive probability” — which every CI statement already carries — is invoked anywhere.

Intersection's $p(x) > 0$ is *global*, and is a genuinely additional hypothesis.

- The proof does not merely condition on a single event with positive probability.
- It takes an equality established under one conditioning event, (w, y, z) with y fixed, and reuses it as though it holds for *every* value of w .

- This includes values of w that could, for a different but still-positive-probability value of y , correspond to a zero-probability triple (w, y, z) .
- Guaranteeing that no such gap exists means *every* combination (x, y, w, z) — not just the conditioning events actually invoked — must have positive probability: a statement about the support of the *entire* joint distribution.
- That is strictly stronger than “the variable I happen to be conditioning on is positive,” which is why it must be stated as a separate hypothesis rather than falling out of the definition of conditioning for free.

In short:

- *Local* positivity ($p(Z) > 0$, $p(W, Z) > 0, \dots$) is built into what it means to write down a conditional probability at all — every rule uses it implicitly, at no extra cost.
- *Global* positivity ($p(x) > 0$ for every joint assignment) is a property of the distribution as a whole, and only Intersection’s proof needs it.
- That is precisely because Intersection is the only proof that reassembles information gathered under two *different* conditioning events into one statement, rather than working within a single conditioning event throughout.

3 The Converses

Does the reverse implication of each rule also hold, making it an *iff*? Two converses hold unconditionally, one holds only under positivity, and one is **false outright** — even under positivity.

Summary of the Converses

- **Symmetry** is trivially an iff: the identical proof runs in both directions.
- **Decomposition**’s converse is **false in general**, with or without positivity: $X \perp Y \mid Z$ and $X \perp W \mid Z$ do *not* imply $X \perp (Y, W) \mid Z$ (XOR counterexample below).
- **Weak Union**’s converse is exactly the forward direction of **Intersection**, which needs $p(x) > 0$. So $X \perp (Y, W) \mid Z \Leftrightarrow X \perp Y \mid (W, Z)$ and $X \perp W \mid (Y, Z)$ is a genuine biconditional *only under positivity*.
- **Contraction**’s converse **holds unconditionally** — it is Weak Union + Decomposition applied together.
- **Intersection**’s converse **also holds unconditionally** — it is Weak Union applied twice. Positivity is needed only to go *up* from two pairwise statements to the joint one, never to come back down.

Proof: Converse of Contraction (unconditional)

Assume $X \perp (Y, W) \mid Z$, i.e. $p(X \mid Y, W, Z) = p(X \mid Z)$.

$$\begin{aligned}
 p(X \mid W, Z) &= \frac{\int p(X, Y, W \mid Z) dY}{p(W \mid Z)} && \text{(product rule)} \\
 &= \frac{\int p(X \mid Y, W, Z) p(Y, W \mid Z) dY}{p(W \mid Z)} && \text{(product rule)} \\
 &= \frac{p(X \mid Z) \int p(Y, W \mid Z) dY}{p(W \mid Z)} && \text{(hypothesis: } p(X \mid Y, W, Z) = p(X \mid Z)) \\
 &= \frac{p(X \mid Z) p(W \mid Z)}{p(W \mid Z)} = p(X \mid Z) && \text{(marginalization)}
 \end{aligned}$$

So $p(X \mid Y, W, Z) = p(X \mid W, Z) = p(X \mid Z)$, i.e. $X \perp Y \mid (W, Z)$ (Weak Union) and $X \perp W \mid Z$ (Decomposition). Neither needs positivity, so this direction of Contraction's biconditional does not either. ■

Proof: Converse of Intersection (unconditional)

Claim: $X \perp (Y, W) \mid Z \implies X \perp Y \mid (W, Z)$ and $X \perp W \mid (Y, Z)$.

This is nothing but **Weak Union** applied twice. Applying Weak Union directly to $X \perp (Y, W) \mid Z$ gives $X \perp Y \mid (W, Z)$. Since (Y, W) is an unordered pair, $X \perp (Y, W) \mid Z$ is the same statement as $X \perp (W, Y) \mid Z$; applying Weak Union to that (swapping the roles of Y and W) gives $X \perp W \mid (Y, Z)$. ■

Counterexample: Decomposition's Converse Is False (XOR)

Take $Z = \emptyset$. Let $Y, W \sim \text{Bernoulli}(0.5)$ independently, and let $X = Y \oplus W$ (XOR). We now work out every marginal and conditional probability needed to verify $X \perp Y$, $X \perp W$, and $X \not\perp (Y, W)$ in full.

Step 1: The joint distribution. Since Y, W are independent fair coins, $p(y, w) = 0.25$ for each of the four (y, w) pairs, and $X = y \oplus w$ is then fixed deterministically — so each of the 8 rows below has probability either 0.25 (if $x = y \oplus w$) or 0 (otherwise):

Y	W	$X = Y \oplus W$	$P(Y, W, X)$
0	0	0	0.25
0	0	1	0
0	1	0	0
0	1	1	0.25
1	0	0	0
1	0	1	0.25
1	1	0	0.25
1	1	1	0

Step 2: Marginal of Y . Sum the joint table over W, X :

$$p(Y=0) = \underbrace{0.25}_{(0,0,0)} + \underbrace{0}_{(0,0,1)} + \underbrace{0}_{(0,1,0)} + \underbrace{0.25}_{(0,1,1)} = 0.5, \quad p(Y=1) = 0 + 0.25 + 0.25 + 0 = 0.5.$$

By the same computation (or by construction), $p(W=0) = p(W=1) = 0.5$.

Step 3: Marginal of X . Sum the joint table over Y, W :

$$p(X=0) = \underbrace{0.25}_{(0,0,0)} + \underbrace{0}_{(0,1,0)} + \underbrace{0}_{(1,0,0)} + \underbrace{0.25}_{(1,1,0)} = 0.5, \quad p(X=1) = 0 + 0.25 + 0.25 + 0 = 0.5.$$

So X is itself a fair coin: $p(X=0) = p(X=1) = 0.5$.

Step 4: Joint of (X, Y) , and $p(X | Y)$. Marginalize the 3-way table over W :

$$p(X=0, Y=0) = \underbrace{0.25}_{W=0} + \underbrace{0}_{W=1} = 0.25, \quad p(X=1, Y=0) = \underbrace{0}_{W=0} + \underbrace{0.25}_{W=1} = 0.25,$$

$$p(X=0, Y=1) = \underbrace{0}_{W=0} + \underbrace{0.25}_{W=1} = 0.25, \quad p(X=1, Y=1) = \underbrace{0.25}_{W=0} + \underbrace{0}_{W=1} = 0.25.$$

Y	X	$p(X, Y)$	$p(Y)$	$p(X Y) = p(X, Y)/p(Y)$
0	0	0.25	0.5	$0.25/0.5 = 0.5$
0	1	0.25	0.5	$0.25/0.5 = 0.5$
1	0	0.25	0.5	$0.25/0.5 = 0.5$
1	1	0.25	0.5	$0.25/0.5 = 0.5$

For every value of Y , $p(X | Y) = 0.5 = p(X)$. Also $p(X, Y) = 0.25 = 0.5 \times 0.5 = p(X)p(Y)$ in every row. Hence $X \perp Y$.

Step 5: Joint of (X, W) , and $p(X | W)$. By the symmetric computation (marginalizing over Y instead of W):

W	X	$p(X, W)$	$p(W)$	$p(X W) = p(X, W)/p(W)$
0	0	0.25	0.5	$0.25/0.5 = 0.5$
0	1	0.25	0.5	$0.25/0.5 = 0.5$
1	0	0.25	0.5	$0.25/0.5 = 0.5$
1	1	0.25	0.5	$0.25/0.5 = 0.5$

Again $p(X | W) = 0.5 = p(X)$ for every value of W , and $p(X, W) = p(X)p(W)$ throughout. Hence $X \perp W$.

Step 6: Joint conditional $p(X | Y, W)$. Now condition on *both* Y and W at once, using the original 3-way table and $p(Y, W) = 0.25$ for every pair:

Y	W	$p(X=0, Y, W)$	$p(X=1, Y, W)$	$p(X Y, W)$
0	0	0.25	0	$p(X=0 0, 0)=1, p(X=1 0, 0)=0$
0	1	0	0.25	$p(X=0 0, 1)=0, p(X=1 0, 1)=1$
1	0	0	0.25	$p(X=0 1, 0)=0, p(X=1 1, 0)=1$
1	1	0.25	0	$p(X=0 1, 1)=1, p(X=1 1, 1)=0$

(Each row divides by $p(Y, W) = 0.25$, e.g. $p(X=0 | Y=0, W=0) = 0.25/0.25 = 1$.) In every row, $p(X | Y, W) \in \{0, 1\}$ — a degenerate (deterministic) distribution — whereas $p(X) = 0.5$. So

$$p(X | Y, W) \neq p(X) \quad \text{for every } (Y, W),$$

i.e. $X \not\perp (Y, W)$.

Conclusion. Steps 4–5 give $X \perp Y$ and $X \perp W$ individually (fully verified above); Step 6 shows $X \not\perp (Y, W)$ jointly (also fully verified). Therefore

$$X \perp Y \text{ and } X \perp W \not\Rightarrow X \perp (Y, W). \blacksquare$$

Intuitively: knowing Y alone (Step 4) or W alone (Step 5) tells you nothing about X , because each is independently a fair coin unrelated to X in isolation. But knowing *both* Y and W

(Step 6) pins X down exactly, since $X = Y \oplus W$. The dependence is entirely a joint, second-order effect that no pairwise marginal can detect — exactly the phenomenon Decomposition’s converse would (wrongly) rule out.

Remark: this table is not strictly positive ($p(y, w, x) = 0$ for half the rows), so it cannot test the positivity-dependent Intersection rule — it only disproves Decomposition’s converse, which fails with or without positivity.

Summary Table

Rule	Forward direction	Converse
1. Symmetry	holds (any p)	holds — identical statement
2. Decomposition	holds (any p)	fails , even if $p(x) > 0$ (XOR example)
3. Weak Union	holds (any p)	= Intersection; needs $p(x) > 0$
4. Contraction	holds (any p)	holds (any p) — Weak Union + Decomposition
5. Intersection	holds, needs $p(x) > 0$	holds (any p) — Weak Union applied twice

Rules 3, 4, 5 are secretly two overlapping biconditionals: Contraction is an unconditional *iff*, and Weak Union/Intersection form an *iff* only under positivity. Positivity does real work in exactly one place — the forward direction of Intersection (equivalently, the converse of Weak Union). Decomposition is the true outlier: its converse fails regardless of any distributional assumption, because a third variable can carry dependence on X that is invisible in Y or W alone but fully exposed once they are considered jointly.